

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SA4 – Pension and Other Benefits

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q. 1)**i)****a) Objective of providing Pension Schemes to employees**

- i. To provide attractive benefits at an affordable and predictable costs/contributions. May prefer to have a stable proportion of budget expenses towards meeting pension cost.
- ii. To attract and retain best talents for government services. A defined Benefit Pension Scheme is a major attraction for people to join the Government Jobs.
- iii. To provide pension benefits which are flexible and comparable with the pay paid while in service (e.g. 40% of pay)
- iv. Consistent with pension benefits being provided to other sections of the society
- v. May wish to use the financial, pension market while delivering pension benefits
- vi. May prefer to change its role i.e. from a provider to a facilitator/educator or from provider to a regulator/developer
- vii. To offer protection to employees in retirement and to the family while in-service death of an employee – paternalistic view

(max 3)

b) To what extent the objectives are met:Under NPS Structure:

- The costs are predictable to the Government. The contributions which are expressed as a proportion of salary will help the government to plan & control it within budgetary provisions.
- The NPS contributions are expressed as a proportion of salary. It means contributions increase with salary escalation in real terms. However, the benefits are not assured and depends upon the investment income during accumulation and prevailing annuity rates at the time of retirement.
- There is no protection to family in case of death of an employee while in-service. The family pension depends upon the accumulated value of corpus.
- A Defined Contribution system may be preferred by people who wants to switch jobs for higher career and who wants control on their retirement savings – An attractive benefit.
- Providing pension benefits based on accumulated corpus is consistent with the pension benefits being provided to other classes of subscribers under NPS.
- The role of Government is gradually shifting from “provider” to “facilitator” (except for the contributions being paid for its employees) The role of benefit provider is being shifted to NPS. The role Governance, records keeping are being given to PFRDA, CRA. In NPS model.
- The pension savings are invested in financial market and which are good for a growing economy like India creating a huge amount of pension assets which can support the economy in the long run.
- NPS model uses the services of insurance companies while providing pension benefits. But the annuities in the markets are not exactly matching the pension benefits/needs of subscribers/pension providers.

Under UPS Structure

- The UPS Structure provides an assured benefit expressed as a proportion of final pay which may be understood and appreciated by the member.
- Benefits offered under UPS are based on final salary; they are similar to the benefits under OPS, which was popular with the Government employees. UPS benefit structure will be attracting (& retaining) best talents towards(in) government jobs.

- This system offers an irrevocable choice to employees between DB or DC. The subscriber can choose a benefit depends upon his/her individual risk appetite and personal circumstances. UPS provides flexibility to members.
- There is no protection to family in case of death of an employee while in-service. The family pension depends upon the accumulated value of corpus.
- The cost of providing the pension benefits based on final salary may be very high & uncertain. This may cause constraints in the budget. However, the accumulated corpus till date provides a cushion to the government in managing the pension scheme effectively as the UPS is a contributory, funded pension system compared to unfunded OPS.
- This new Defined Benefit Scheme is offered only to Central Government Employees and the Corporate and individual members models are still under Defined Contribution Model only and no assured benefits to these models.
- The government is just playing the role of principal sponsor & the role of payment of benefits is being shifted from insurer to NPS.

(Max 8)

ii)

a) The choice is depending upon the circumstances of the individual and the following information are required from the employee to assess the individual circumstances.

1. Potential future service, prospects of future salary growth
2. Investment option chosen & how it will vary in future
3. Pension/lump sum needs at retirement
4. Financial awareness of the individual
5. Family status of the individual (single/married)
6. Type of regular income he prefers (fixed or increasing regular income)
7. Tax position of the individual member
8. Other sources of income after retirement
9. Attitude towards risk

(Max 3)

b) One simple approach will be to project the present corpus of Rs. 1.25 crores up to NRA 60 allowing for individual member contributions together with Government contributions.

The projection is to be carried out using salary growth, investment return relevant to the individual.

The projection of final salary for UPS benefit should take into account both general salary & promotional growth assumptions.

It may be reasonable to assume that he survives up to NRA while comparing retirement benefits.

Actuary needs to estimate the pension (i.e. similar to UPS) the accumulated corpus under NPS will buy at the time of retirement using the appropriate annuity rates prevailing at the time of retirement.

The annuity expressed as a % of final salary under NPS can be compared with benefits available under UPS.

If there are difficulties in finding annuities that exactly matches with the UPS benefits, suitable adjustments are to be done in the calculations.

(Max 4)

c) Results are to be presented to the client in such a way to help him take informed decision while exercising the option as the client is taking an irrevocable decision.

Results should be presented in accordance with the provisions of APS – 27 issued by the Institute of Actuaries of India.

May specify the relevance of assumptions in the projections & the actual benefits will differ as compared to illustrations.

May illustrate the pensions members may get using the projected (present) corpus of Rs. 1.25 crores without future contributions & compare with accrued pension under UPS.

May illustrate the revised (reduced) benefits in case the members commute portion of pension at retirement.

May illustrate the reduced benefits if he chooses to retire at an earlier date.

The risk of choosing an aggressive investment option under NPS can be expressed in terms of additional contributions if such investment option brings lower investment return as compared to bench mark returns.

The variability of illustrated benefits due to changes in the annuity market conditions should be presented with several scenarios.

It should be clearly mentioned that changes in the tax position, risk profile of member may significantly alter comparisons.

(Max 4)

d) APS 27 provisions (10.3)

- Particular care should be taken in preparing an illustration that will be used to compare defined benefits and benefits from defined contributions, since it is possible that scheme beneficiaries will make an irrevocable choice of future benefit arrangements based on such an illustration.
- Sufficient information should be provided to enable the scheme beneficiaries to understand the difference between defined benefit and defined contribution as regards the relative risk borne by the employer and the employee. The assumptions used to project the defined benefit and defined contribution benefits should be consistent.
- An illustration prepared for this purpose should show separately the illustrated benefits from existing assets and the illustrated benefits from future contributions, if any, so the recipient can compare both current accrued benefits and ultimate retirement benefits assuming service continues to the illustrated retirement age.
- It should be ensured that the illustration draws the scheme beneficiaries' attention to other benefits that may be advantageous to that beneficiary, and which differ significantly between the arrangements being compared.

(Max 3)

iii) The primary risk to the NPS is that the (true) cost of providing the pension benefits is more than the value of accumulated contributions. Since UPS is an ongoing scheme, it is not possible to measure the risk with reasonable confidence until it actually occurs.

These risks faced by the NPS Trust may relate to one or more of the following risks:

1. Investment related risks – ALM, Reinvestment, Investment Strategy, Liquidity, Credit risks and concentration risks
2. Demographic risks – Mortality & Attrition during Employment and Longevity after retirement
3. Financial Risks – Salary Escalation, Pension Increase, Expense and inflation risk
4. Operational risks – coordination between various departments and institutions, data, incorrect payments and mismanagement

5. Specific risks related to scale and size of the scheme – Investments, Payments, Data Management, etc

How a mandatory actuarial investigation will help the Trust in managing these risks:

When the UPS is in operation, UPS fund will be exposed to the several risks as stated earlier due to the uncertain nature of benefits, contributions, investment income etc.

Actuarial investigation on periodic basis helps to assess the adequacy of funds for paying the benefits taking into account the various risks relating to the scheme.

In that process, Actuary will be makes assumptions on several key financial, demographic parameters and makes a prudent estimate of liabilities & assets to check the adequacy.

The process involves several key steps:

- validating the current year's member data
- reconciling this data with that of the previous year
- conducting experience analyses
- setting assumptions
- estimating liabilities
- analysing surpluses
- Presenting the results
- and preparing various disclosures

Each step uncovers specific risks impacting the UPS fund. For instance, data validation and reconciliation highlight data issues like missing, duplicate, inconsistent, or incorrect data thus helping the Trust to manage the operational risks.

Experience analysis evaluates the relevance of existing assumptions & its consistency with the financial market conditions & demographics specific to the scheme, identifies deviations, and examines the underlying causes, thereby identifying and mitigating demographic and financial risks.

The insights gained through actuarial investigations play a vital role in asset-liability management, helping to mitigate liquidity and reinvestment risks effectively.

The surplus analysis helps the Trust to identify the causes for the deviations from the expected results related to demographic and financial risks so that the Trust can take corrective action in-time.

Disclosures include surplus analysis, liability reconciliation, sensitivity analysis, duration-based liability breakdowns, and immediate contribution requirements. These disclosures allow NPS& other stake holders t to understand the diverse risks such as Investment Risks, Credit Risks and specific risks faced by the UPS

Several stakeholders of UPS need a value of the benefits being provided under UPS. Ex: NPS trust may need such value for accounting purpose; subscribers to understand the cost. Actuarial investigation helps to place a value on UPS benefits being provided taking into account various risks.

(max 10)

iv) Method used to value the liabilities:

The purpose of valuation will be to

- Arrive at a Liability so that the gap funding requirement can be assessed.
- Arrive at a schedule of contributions to meet the future shortfall.

The valuation has to be done separately for the employees and the pensioners. To value the liabilities of employees covered under the UPS, the Projected Unit Credit Method can be used.

The accrued benefit of each member will be projected up to retirement age using salary growth assumptions & discounted to the present time using suitable discount rate.

The liabilities pensioners including family pensioners are estimated using discounted cash flow approach, using assumptions on pension increases, longevity etc

Total value of the liabilities of in-service members & pensioners less the market value of the assets will be the gap funding requirements.

Gap funding will be amortized over a future period (eg future service) to arrive at the schedule of contributions.

How the assumptions can be arrived:

The assumptions for any valuation depend on the purpose of the valuation and should be appropriate for the purpose. Assumptions should be derived using relevant and sufficient data.

The valuation will involve use of long-term assumptions. These assumptions should be mutually consistent and indicative of future expectations for the underlying employee demographics.

The following are the list of assumptions involved in the valuation exercise:

Discount Rate:

Discount Rate should be based on Current investment pattern of assets, Current investment yield, expected investment yield on the existing investments and the expected reinvestment rate for the new money. If any inflation assumption is used for pension increase or expenses then the inflation rate should be consistent with the discount rate.

Salary Escalation Rate

The salary escalation rate will be based on the pay scales mentioned in the latest Pay Commission Report considering promotional and expected inflation increases. The inflation rate used will be consistent with the inflation rate used in expense and pension increase rate.

Pension Increase Rate

As the scheme is providing dearness relief to pensioners, a pension increase rate based on the recent data and the expected inflation rate should be arrived.

Expenses:

For valuation purposes, the expenses relating to the scheme can be expressed either as a cost per pensioner or as a proportion of the total pension payments in effect.

Pre-Retirement Mortality Rate:

The Mortality Rate should be based on scheme data if it is sufficient. Otherwise, the latest Standard tables published by Institute of Actuaries of India with appropriate discounts/loadings can be used.

Post-Retirement Mortality Rate:

The Mortality Rate should be based on scheme data if it is sufficient or should be based on the latest annuitants' mortality table published by Institute of Actuaries of India. The

increase in longevity when compared to the applied mortality table due to time gap and the future increase in longevity also should be factored in the projection.

Attrition Rate:

The Attrition Rate should be based on scheme data if it sufficient. Otherwise, industry level or age/service-based attrition can be used.

Other assumptions:

Other assumptions may include the proportion of peoples availing lump sum benefits at retirement, spouse age if not given in data, etc

(Max 10)

v. The following information about the UPS can be provided to employees and pensioners on an annual basis:

- Summary of employees and pensioners data – Number of employees, Number of pensioners, Average Age of employees, average age of pensioners and average salary/pension, etc
- Summary of investments – Investment return of individual accounts for the period, Investment return of pooled fund, Investment patterns, Assets split by major categories, credit risks faced by the portfolio during the period, Number of pension funds, AUM for each Pension Funds
- NPS may disclose its policy in managing the funds. It may cover area investment ,payment of benefits, Grievance redressal mechanisms etc.
- Contribution paid by the members and Government separately disclosing any additional contribution made by the Government
- Total pension payouts, Dearness relief granted during the period and expenses charges by Pension Funds and Pooled Corpus
- Total Assets and total Liabilities of the UPS valued by actuaries and the extent of gap or funded status and comparable figures of previous year
- Any major changes in Rules/Regulations/Contributions/Benefits/Member Profile/Asset Profile/Investment Strategy related to UPS related to the period
- General description of risks faced by the UPS and the mitigation techniques applied to manage/reduce those risks
- Any significant event that have implications on the fund (e.g. court judgement, financial fraud)
- May provide the names of professional involved in the management of the fund e.g Board of Trustees, Actuaries, Fund Manager, Auditor etc.

(Max 5)

[50 Marks]

Q. 2)

i) Reasons for initiating a communication by Actuary X:

To understand the Gratuity Scheme of the company; especially the benefits of the scheme, profile of members, funding status & financial significance of the benefits in company's accounts, administrative mechanisms for paying benefits& complying with various laws.

To understand the approach of the previous actuary in valuing the gratuity benefits especially with reference to valuing benefits payable on different contingencies, developing basis for various parameters & the projection models used by him in deriving results.

The meeting will help him to get the professional guidance & expert opinion of the previous actuary.

To get insight into features of the scheme specific to the company for eg unpaid contributions, self-investment details if any, discretionary benefits provided if any, legal disputes relating to the benefits.

This will help him such significant factors are not missed in his valuation work.

It is mandatory for Actuary X, as per PCS standards & Actuary Act 2006 to ascertain the professional reasons to be considered while accepting the appointment. From the previous actuary.

This initiative will result in best professional advice to clients & upholding the high practice standards set by the profession.

(Max 4)

ii) Data/information/reports to be called for the valuation

Members data such as salary roll number, designation, DOB, Date of joining the scheme, Pay & other allowance for the purpose of the scheme as on 31.03.2025

Details of benefits provided on various contingencies such as retirement, death, early retirement, resignation, eligibility conditions, retirement age etc

Data relating to assets of the scheme- schedule of assets & its fair value as on 31.03.2025

Details of cash flow such as contributions paid, benefits paid for the previous (one or more) years

Any significant events since the last actuarial valuation eg merger/acquisition activity, VRS initiative, mass recruitment, changes in benefits Legal disputes relating to the scheme

Previous Actuarial valuation reports of the scheme, audited financial statements of the company, minutes of Trustee meetings, Board resolutions relevant to the scheme.

(Max 4)

iii) Actuarial approach to calculate the value of benefits & assets as on 31.03.2025

Objective:

The aim of the approach of the actuary X is to calculate the present value of the accrued benefits (PVAB) of all members allowing for salary growth, death & other decrements as per scheme rules as on 31.03.2025.

He needs to ensure that the arrived figures are not lower than the corresponding figures calculated as per Gratuity Act 1972.

Data:

Full members data has been provided for valuation. It means details regarding DOB, DOJ, Salary as on 31.03.2025 (i.e. basic pay, Basic+DA) are available in respect of all members.

Before using the data, actuary X has to ensure the consistency & sufficiency of the data. For eg he can compare the average age, average past service, average basic, average (Basic+DA) with the last year figures.

A reconciliation of the member numbers between the last year & the current year may be done to ensure the reasonableness of data.

Developing model:

The cashflow model should calculate the accrued benefit of each member allowing for salary growth upto Normal retirement.

PU method should be used in the calculations.

Model should generate benefit cash flows for death, early exit, & retirement for a given set of parameters.

Model should be capable of capturing the higher death benefits, maximum ceiling of the gratuity plan & also the ceiling under Gratuity Act.

The cash flows should be discounted to the present time to arrive at the value of the accrued benefits.

The model should calculate the present value of the accrued benefits both as per scheme rules & as per Gratuity Act.

Initially, the parameters used in the last year valuation may be used in the projection to verify the reasonableness of the results.

Changes in the discount rate, salary growth & other decrements may be considered taking into account the changes financial, demographic environment.

The changes in the basis should conform to the provisions of APS27 set by the Profession & Ind As (19)

Model should be re-run using the revised set of parameters and the value of the accrued benefits under both rules (as per scheme rules & as per gratuity Act 1972) should be calculated for each member.

The higher value of the projected accrued benefit as on 31.03.2025 should be summed across all members to generated PVAB of the scheme.

Fair value of all assets as at 31.03.2025 as provided by the employer may be compared with last year figures taking into account the benefits paid, contribution.

The investment return earned on each class of assets may be compared with Government, corporate, equity yield prevailing to ensure its reasonableness.

Presentation of results:

Present value of the accrued benefits (PVAB) less the fair value of the plan assets gives the funded status of the scheme as at 31.03.2025.

The variability of the results may be illustrated for several set of consistent parameters.

Guidelines may also be provided on next year gratuity cost, contributions depending on company's requirements.

Figures to be reflected in Balance sheet, Profit in loss account, Other Comprehensive Income (OCI) may be provided separately to fit Ind AS (19) requirements.

(Max 12)

iv) Plan amendment & curtailment

A plan amendment occurs when a company introduces, or withdraws, a defined benefit plan or changes the benefits payable under an existing defined benefit plan (for example Gratuity benefit).

A curtailment occurs when an entity significantly reduces the number of employees covered by a plan. A curtailment may arise from an isolated event, such as the closing of a plant, discontinuance of an operation or termination or suspension of a plan. (example specially designed VRS)

Past service cost may increase or decrease due to the plan amendment or curtailment.

Changes in the past service benefits are to be recognized in the profit loss account separately for plan amendment & curtailment when such plan amendment occurs (or when the curtailment plan is introduced).

Settlement

A settlement occurs when the company settles the benefit or transfers the legal obligation to another company.

The gain or loss on a settlement is the difference between: (a) the present value of the defined benefit obligation being settled, as determined on the date of settlement; and (b) the settlement price,

The company shall recognize a gain or loss on the settlement of a defined benefit plan in the profit & loss account as settlement gain/loss when the settlement occurs.

Qualifying Insurance policy:

A qualifying insurance policy is an insurance policy issued by an insurer that is not a related party (as defined in Ind AS 24, Related Party Disclosures) of the company,

if the proceeds of the policy: (a) can be used only to pay or fund employee benefits under a defined benefit plan; and (b) are not available to the reporting entity's own creditors (even in bankruptcy) and cannot be paid back to the company

They are part of plan assets. While computing the net defined benefit liability (ie surplus or deficit) they can be deducted along with other plan assets at its fair value.

Reimbursement

Right to reimbursement arises if there exist another source (e.g. parent company) available to the company to pay part or all of the expenditure required to settle a defined benefit obligation.

Ind As19 considers an un-qualifying insurance policy as a "right to reimbursement"

Right to reimbursement is to be measured at its fair value and shall be shown separately in the assets.

A brief description of the link between the reimbursement right and the related obligation should be disclosed.

(Max 11)

v)

a) Possible reasons for the deficit:

There is an opening deficit of 72.41% and it is growing. Poor funding is the primary cause for the deficit

Liability figures increase from Rs.8700 (lakhs) to 9300 (lakhs) over the year. But there is a corresponding increase of Rs100(lakhs) on the asset side. This indicate either lower investment return or contributions are not paid at recommended rates.

Maturing profile of members in terms of increasing past service, increasing average age. Higher proportion of the members may be approaching maximum accruing benefits.

Increase in number of members by 1000.The profile of members joining the scheme may be a source. Example: Any new entrants with considerable past service without appropriate asset transfer by merger activity.

Actual experience Vs assumed

More number of deaths than that assumed & the profile of deceased members

Less withdrawal than that assumed especially at younger ages without satisfying eligibility criteria if any.

But we have no further information to analyse the experience.

Strengthening financial parameters discount rate, to the extent of 0.25% will also be a source for the deficit.

Other factors: incorrect payment, data error, modelling error

(Max 5)

b) issues to be discussed before insuring the death benefits.

The company has to analyse the past claim experience. 3 or more years are to be considered for this exercise. Analysis is to be done in terms of age, service, employee category etc.

This analysis will provide information about the significance of the risk, factors contributing to the risk & strain it is causing on gratuity fund.

Large levels of death at younger ages with few years of service causes strain on the fund whereas higher deaths at higher ages (with higher service) will cause less strain on the benefits.

Insuring the death benefits is a good risk mitigation initiative.

It makes sense to insure only the prospective benefits for each employee. ie sum at risk for each member will be "30 months salary-accrued past service benefits" - This sum at risk will keep on reducing as members service increase over the years.

The Group One Year Renewable Term Insurance offered by insurers provides flexibility to change the sum at risk for each member every year.

The premium will be charged for each of the member will be calculated as sum at risk x age related premium and summed across all members.

Higher death risk coupled with lower premium at younger ages (& vice versa) will help the company to get stable rates year after year.

But the insurer will be loading for his expenses, profit & taxes. Therefore, the premium will be higher than the cost of cover if un insured.

But competition, profit sharing feature will limit the insurer from excess charging.

By insuring the death benefits, death risk & the volatility associated with it transferred to the professional manager at a little higher cost & the company gets the benefit of stability.

But the financial status of the scheme is unlikely to be altered significantly by this proposal as death risk is not the primary cause of the deficit.

(Max 6)

c) impact of reducing the Gratuity to benefits to 25 months Pay.

Reducing the benefit, in general, will reduce the cost and hence improve the financial status of the scheme.

But this proposal may cause resentment among the work force. If the "cut" is specific to the company A, it may lose competitive edge in the industry. It is likely to face challenges in recruiting & retaining young talent.

The reduction in benefits is also limited by the minimum stipulated as per Gratuity Act 1972.

It is easier to implement the reduction prospectively. Existing members are not impacted by the proposal.

But there will be no immediate improvement in the financial status of the scheme. The impact of reduction will be recognized in the financials of the scheme as & when the new members replace the existing members.

If the company considers the reduction of (accrued) benefits of existing members, there will be immediate reduction in the present value of Accrued benefits (PVAB).

But existing members will be adversely impacted by the proposals. Eg members who have already reached the maximum (30 months) will have immediate & direct reduction in their retirement benefits. Other members are also likely to feel the impact of the reduction as they complete more number of years of service.

Company A is likely to face legal, compliance challenges to implement the proposal.

Alternatively, Company may consider gradual or stepped -up reduction to the benefits. E.g. Members within 5 years to retirement – no reduction; 10-6 years 29 months pay; 15 to 11 years 28 months etc

The proposal to reduce the benefits will reduce the cost but unlikely to solve the problems (of increasing deficit) of the company.

Company may consider other alternatives

Immediate cash injection to reduce the deficit and make contribution at the recommended rates. Due to the strengthened funding approach, there will be opportunity cost to the company. But the company benefits from the reduction in the net interest cost as the investment from the scheme assets grow.

Leave the deficit as it is in the balance sheet of the company. The company is under no legal obligation to fund the gratuity liability. As long as the covenant of the company is strong, & the financial challenge of the employer is temporary, there will be no immediate threat to the security of the vested benefits of members.

(Max 8)

[50 Marks]
